

Syed Muhammad Hashir

Computer Engineer

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PROFESSIONAL SUMMARY

Computer Engineering graduate with hands-on experience spanning embedded systems, software development, and cybersecurity. Background includes end-to-end embedded product development (Embedded Linux, firmware, hardware-software integration), offensive security work (web application exploitation, password cracking, malware analysis and detection labs), and technical mentorship — currently leading a pod of interns at a research centre. Comfortable working across programming, hardware, and security tooling, with a consistent track record of hands-on teaching, workshop delivery, and team leadership.

SKILLS

Programming: Embedded C, C, Python, Java, JavaScript, PHP, SQL, HTML/CSS, PIC18 Assembly, Arduino C, MATLAB, Verilog

Embedded Platforms & Protocols: Raspberry Pi, Arduino Uno, ESP32, ESP8266/ESP01, PIC18, MSP430, NodeMCU, UART, I2C, SPI, PWM, GPIO, Wi-Fi (802.11)

Security Tools: Burp Suite, Wireshark, Kali Linux, Metasploit, Nmap, Splunk, Sysmon, John the Ripper, Hashcat, FlareVM, REMnux

Security Concepts: SQL Injection, Cross-Site Scripting (XSS), Phishing Analysis, Brute Force Attacks, Password Cracking, Network Reconnaissance, Malware Analysis & Detection

Platforms & Tools: Embedded Linux, Git/GitHub, MPLAB, Arduino IDE, LTSpice, Proteus, SimulIDE, VirtualBox, Cisco Packet Tracer

WORK EXPERIENCE

Embedded Systems Engineer - *Centre of Intelligent Systems and Network Research (CISNR)* Dec 2025 - Present

- Designing and building a portable battery-powered Early Microbial Detection Kit for field-deployable water quality analysis.
- Automated the image capture pipeline so the device autonomously captures sample images at set intervals, with no manual intervention or repositioning required.
- Switched sample containment to transparent glass petri dishes and resolved condensation issues that occurred during bacterial incubation, maintaining consistent image clarity.
- Consolidated control of all components i.e. sensors, motors, fans, heaters, camera, and lighting onto a single Arduino Uno, and migrated the platform's controller from Raspberry Pi to a compact portable PC running a custom webapp for full device control and monitoring.
- Developed control logic for RGB LED arrays to ensure consistent lighting conditions during sample image capture.
- Diagnosed and resolved hardware-software integration issues involving GPIO-controlled peripherals, improving system stability during long-duration testing.
- Serving as pod lead, directly supervising and mentoring a pod of about 10 interns.

Embedded Systems Intern - *National Centre for Artificial Intelligence (NCAI)* Jan 2025 – Feb 2025

- Designed and delivered an automated plant monitoring and control system integrating environmental sensors and Wi-Fi-based remote actuation.
- Implemented real-time monitoring using DHT11 and soil moisture sensors with threshold-based autonomous control logic.
- Built a web-enabled remote control interface supporting both automated operation and manual override.

Independent Security Researcher - *PortSwigger Web Security Academy* 2026

- Completed 23 progressive labs (12 Apprentice, 11 Practitioner) exploiting real-world vulnerabilities across SQL Injection and Cross-Site Scripting attack surfaces.
- Used Burp Suite Repeater and Intruder to manipulate HTTP requests, fuzz input fields, and automate parameter-level attacks.

Security Awareness Analyst (Virtual Experience) - *Mastercard Cybersecurity, Forge* April 2026

- Analyzed a simulated phishing campaign, identifying social engineering techniques and structural patterns used to target corporate employees.

- Submitted a written vulnerability assessment and proposed role-specific training interventions for high-risk teams based on observed threat behavior.

PROJECTS

Malware Analysis & Detection Home Lab:	2026
• Built a malware analysis lab using FlareVM (Windows analysis target) and REMnux (Linux network simulator) on an isolated VMware Host-Only network. • Resolved a DNS service failure in REMnux's INetSim by writing a custom Python UDP socket DNS sinkhole server, and automated lab startup with a bash script. • Separately built a Splunk/Sysmon detection lab: simulated a reverse-TCP malware attack, captured it via detection tooling, and configured and verified a custom detection alert.	
SHA-256 Password Cracking Tool:	2026
• Recovered weak credentials from unsalted SHA-256 hashes using dictionary-based attacks, validating the tool against a controlled test set. • Benchmarked John the Ripper against Hashcat, documenting secure password storage practices including salting, key stretching, and adaptive hashing.	
SSH & Web Login Brute Force Tools:	2026
• Automated credential attacks against SSH authentication and web login forms using custom wordlists and username enumeration. • Documented defensive mitigations including account lockout thresholds, fail2ban configuration, rate limiting, and CAPTCHA integration.	
Detection of Spoiled Food using CNN:	2024
• Developed a CNN-based image classification model using TensorFlow/Keras to distinguish fresh and spoiled food images. • Preprocessed real-world datasets and achieved about 98% test accuracy using Python, NumPy, and image augmentation techniques.	
Home Automation Light System:	2024
• Designed a lighting controller using a PIC18 microcontroller, IR sensors, and interrupt-driven Assembly firmware. • Implemented direct register-level programming and validated the complete system through Proteus simulation.	
Smart Car Parking:	2022
• Built an automated parking management system using Arduino-based Embedded C and a real-time p5.js graphical interface.	
RFID Access Control System:	2022
• Designed and demonstrated an RFID-based physical access control system using microcontroller integration and card authentication logic; awarded 1st place at a university-level security exhibition.	

CERTIFICATIONS

- Google Cybersecurity Certificate (March 2025)
- Python 101 for Hackers — TCM Security (March 2026)
- Mastercard Cybersecurity Virtual Experience — Forage (April 2026)

EDUCATION

B.S Computer Engineering, Comsats University Islamabad, Abbottabad Campus [CGPA 3.75/4.0]	2021 - 2025
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ACHIEVEMENTS & LEADERSHIP

- Pod Lead at CISNR mentoring and supervising a pod of about 10 interns.
- 1st Place in InnovatEx 2025 Project Expo.
- 1st Place in World Telecommunication & Information Technology Day Exhibition.
- Vice President of IET OnCampus CUI Abbottabad.
- Resource Person in IEEE CUI Abbottabad IoT Projects Development Workshop.
- Graphics Head of Robotics Society CUI Abbottabad (2022–2024).

LANGUAGES: English (Professional), Urdu (Native), Pashto (Native)